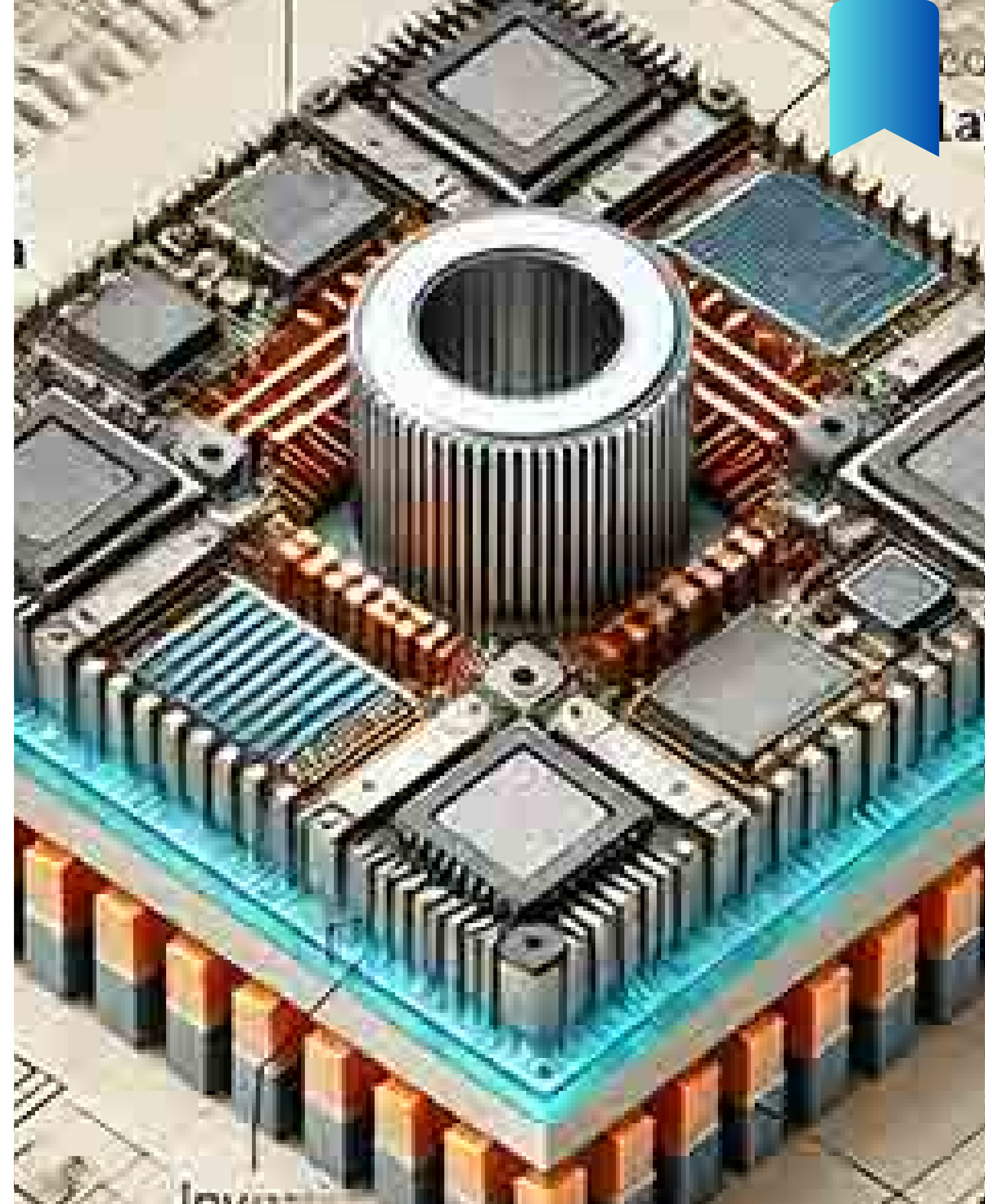


TSV Design Equations & HSpice Simulation



By: Kamineni Dhana Sri Keerthi

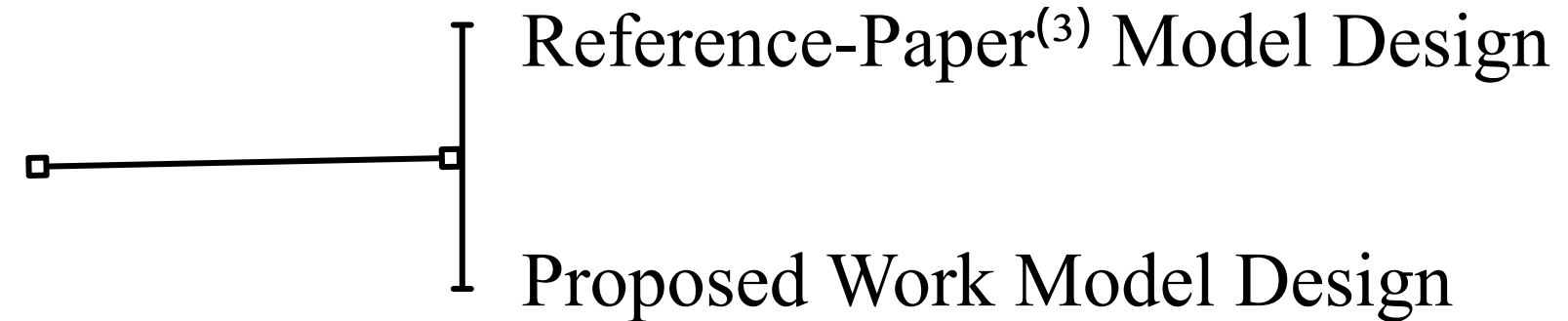


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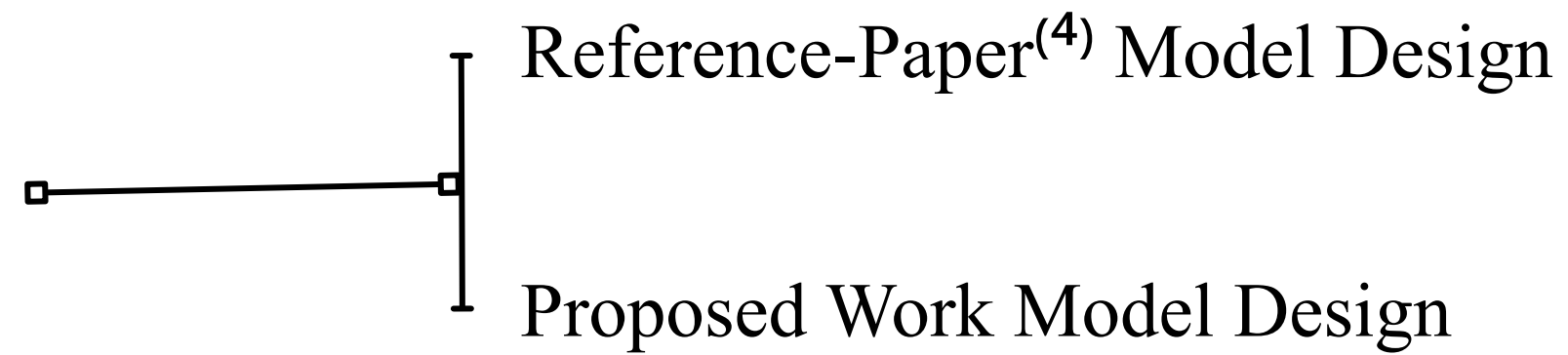
TSV Models

01 Defect Free TSV

02 Defected-TSV In Terms of
Rsht and Lsht

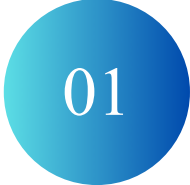


03 Defected-TSV In Terms of
Cleak and Gleak



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TSV Model

-  01 TSV Structure with Equivalent Electrical Model.
-  02 Derived Equations for the Electrical Components (R, L,G, C).
-  03 Hspice Simulation (Calculation of Rise Time and Fall Time)

Defect-Free TSV

Equivalent Circuit

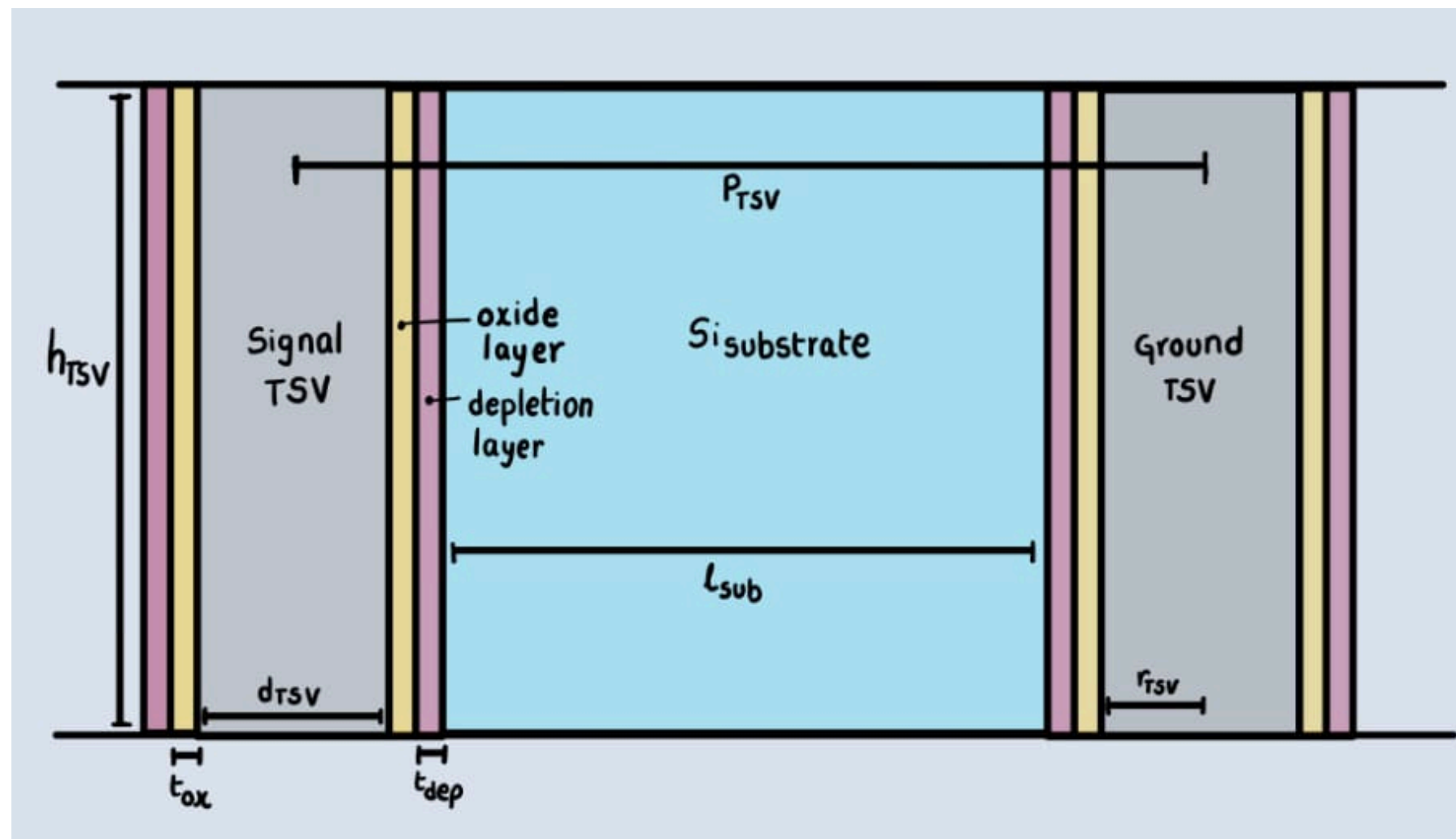


Fig.1.a Defect free TSV
Physical Structure

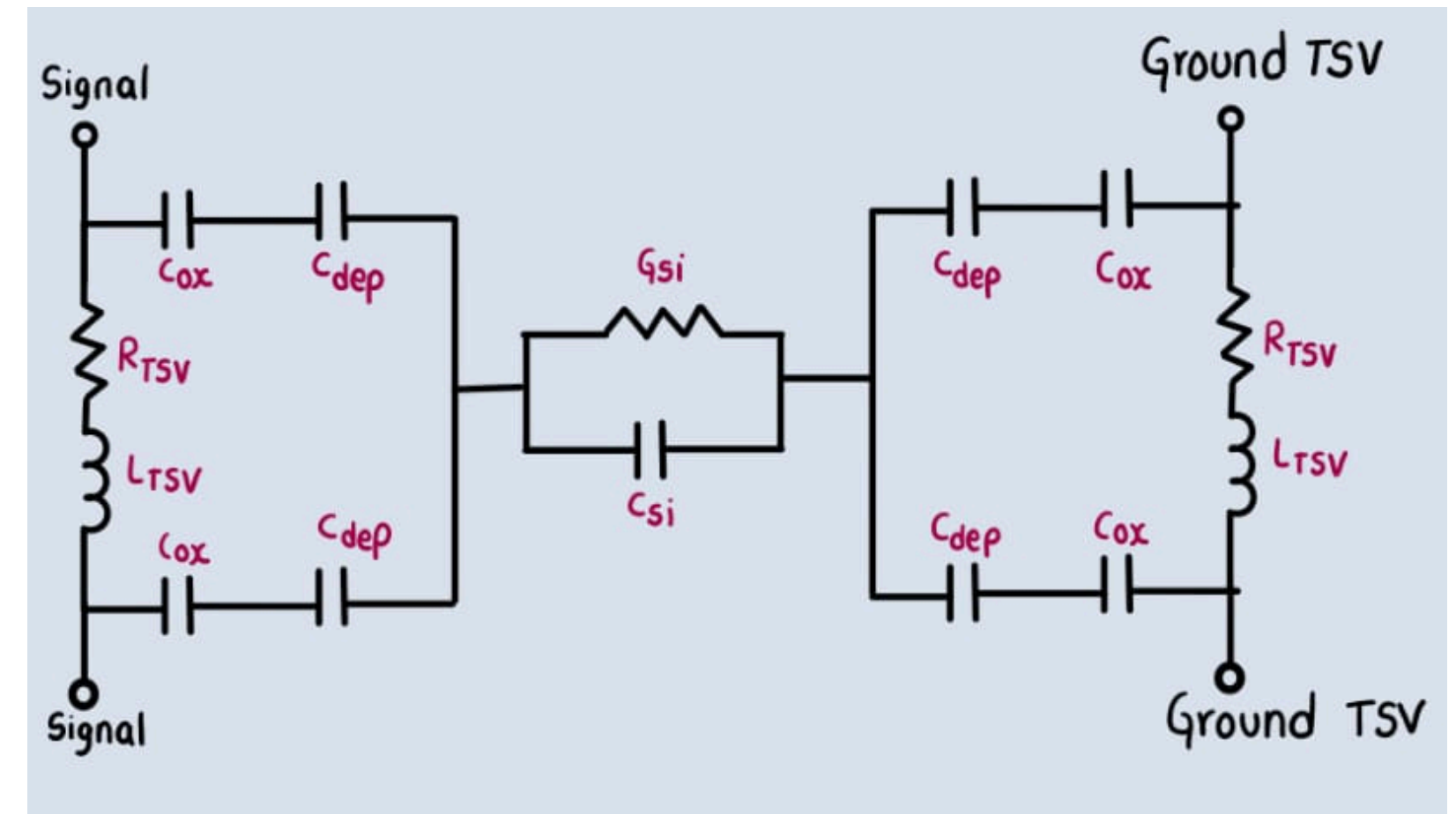


Fig.1.b Defect Free TSV
Equivalent Electrical Model

Table

Table.2.a Formula terms Table

Symbol	Meaning
pTSV	Pitch between two TSV's[=4rTSV]
dTSV	Diameter of TSV
rTSV	Radius of TSV
hTSV	height of TSV
tox	Thickness of oxide layer
tdep	Thickness of Depletion layer
l substrate	length of the Substrate
σ_{si}	Conductivity of Silicon Substrate
σ_{cu}	Conductivity of copper
ρ_{TSV}	Resistivity of TSV
$\epsilon_{r,si}$	Relative Permittivity of Silicon
$\epsilon_{r,ox}$	Relative permittivity of oxide layer
$\mu_{r,TSV}$	Relative permeability of TSV
δ_{ox}	loss tangent of oxide layer
δ_{si}	loss tangent of Silicon

Table.2.b Formula terms Table

Symbol	Meaning
J	Leakage Factor
θ	Angle of the Sector [Pinhole]
Lp	Length of the Sector [Pinhole]
ρ_T	Resistivity of the material at temperature
T	The temperature at which you want to find the resistivity
T0	Reference temperature
α	Temperature coefficient of resistivity
Lp	Length of the sector [pinhole]
ρ_0	Resistivity at the reference temperature
ρ_T	Resistivity of the material at temperature
θ	Angle of the Sector [pinhole]

Defect-Free TSV

Equations {R, L, G, C}

$$R_{TSV} = \sqrt{R_{dc}^2 + R_{ac}^2} [ohms] \quad \dots Eq1$$

Where

$$R_{dc} = \frac{\rho_{TSV} h_{TSV}}{\pi (r_{TSV})^2} [ohms] \quad \dots Eq2$$

$$R_{ac} = \frac{\rho_{TSV} h_{TSV}}{2\pi r_{TSV} \delta_{sd} - \pi \delta_{sd}^2} [ohms] \quad \dots Eq3$$

$$\delta_{sd} = \frac{1}{\sqrt{\pi f \mu_{TSV} \sigma_{TSV}}} [m] \quad \dots Eq4$$

Defect-Free TSV

Equations $\{R, L, G, C\}$

$$L_{TSV} = \left[\frac{1}{2} \frac{\mu_{TSV} h_{TSV}}{2\pi} \left[\ln \frac{\rho_{TSV}}{r_{TSV}} \right] \right] [H] \quad \dots \text{Eq5}$$

$$G_{si} = \frac{\pi \sigma_{si} h_{TSV}}{\cosh^{-1} \left[\frac{\rho_{TSV}}{d_{TSV}} \right]} [s] \quad \dots \text{Eq6}$$

$$C_{ox} = \frac{2\pi \epsilon_{ox} [h_{TSV}]}{\ln \left[\frac{r_{TSV} + tox + tdep}{r_{TSV} + tox} \right]} [F] \quad \dots \text{Eq8}$$

$$C_{dep} = \frac{2\pi \epsilon_{si} [h_{TSV}]}{\ln \left[\frac{r_{TSV} + tox + tdep}{r_{TSV} + tox} \right]} [F] \quad \dots \text{Eq9}$$

$$C_{si} = \frac{\pi \epsilon_{si} [h_{TSV}]}{\cosh^{-1} \left[\frac{\rho_{TSV}}{d_{TSV}} \right]} [F] \quad \dots \text{Eq10}$$

$$C_{ins} = \frac{C_{dep} C_{ox}}{C_{dep} + C_{ox}} [F] \quad \dots \text{Eq11}$$

Defect-Free TSV

H-Spice Simulation

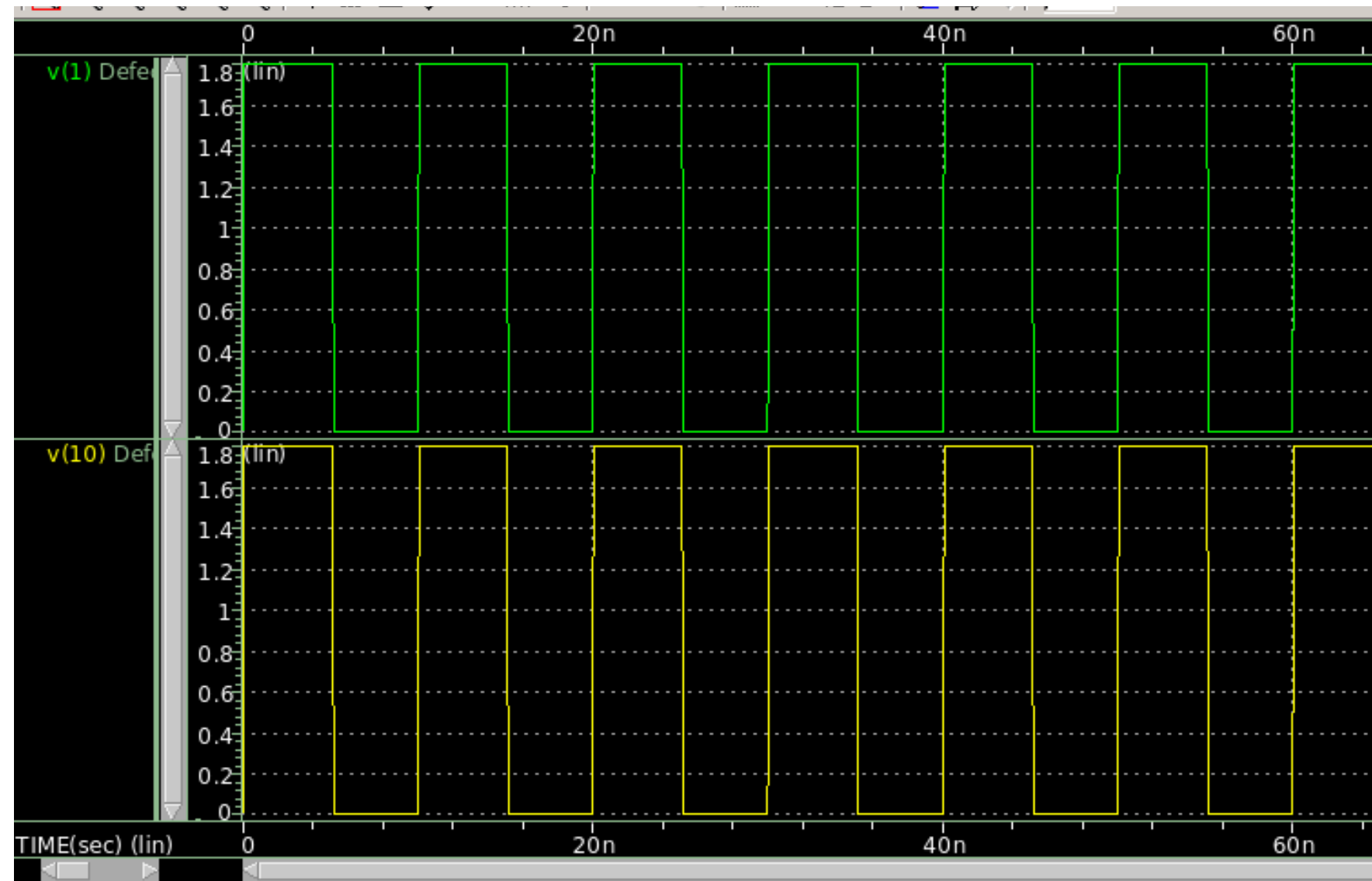


Fig.3.a Input voltage and output voltage

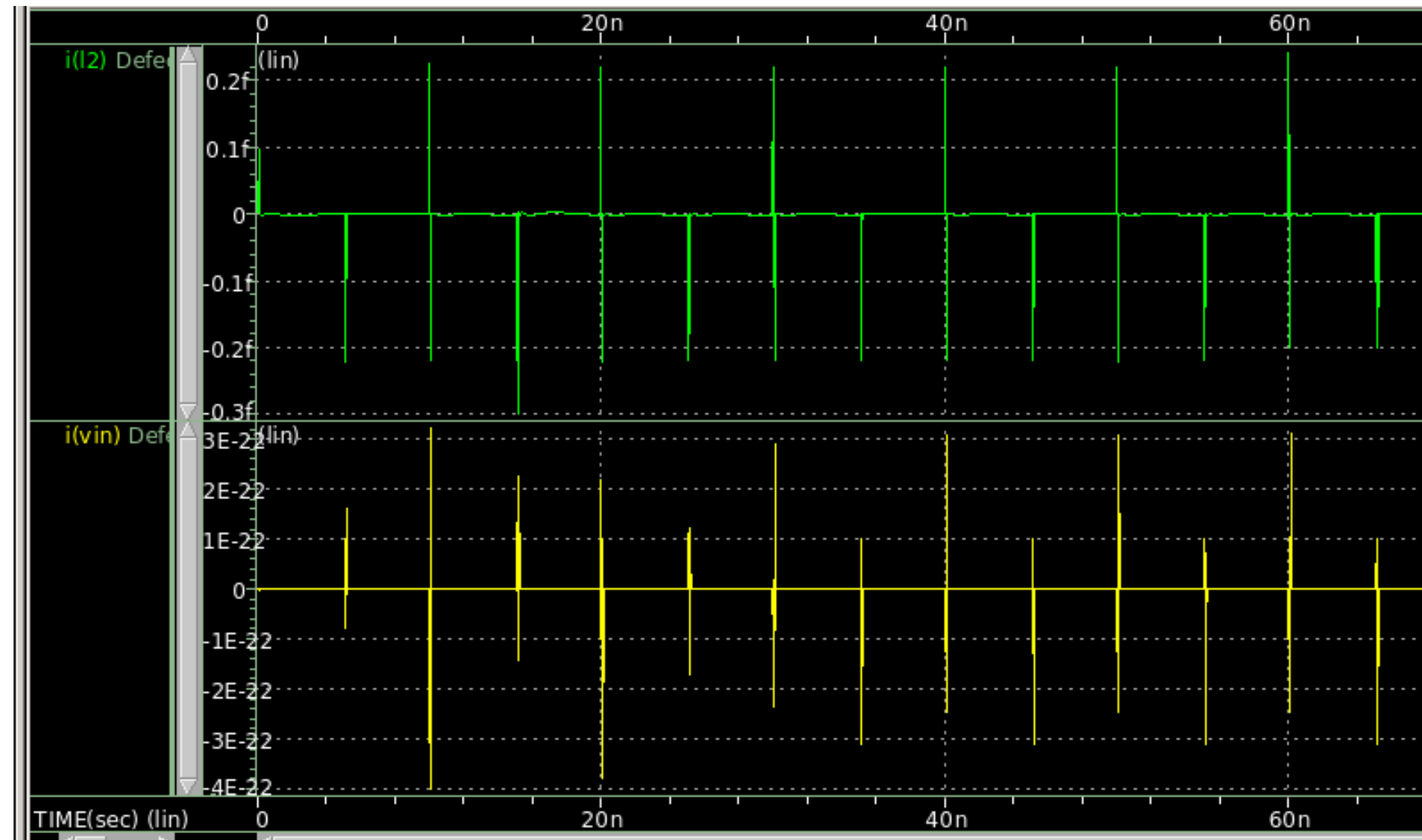


Fig.3.b Input current and output current

Pinhole Defected Coupled via (Rsht and Lsht)

Equivalent Circuit

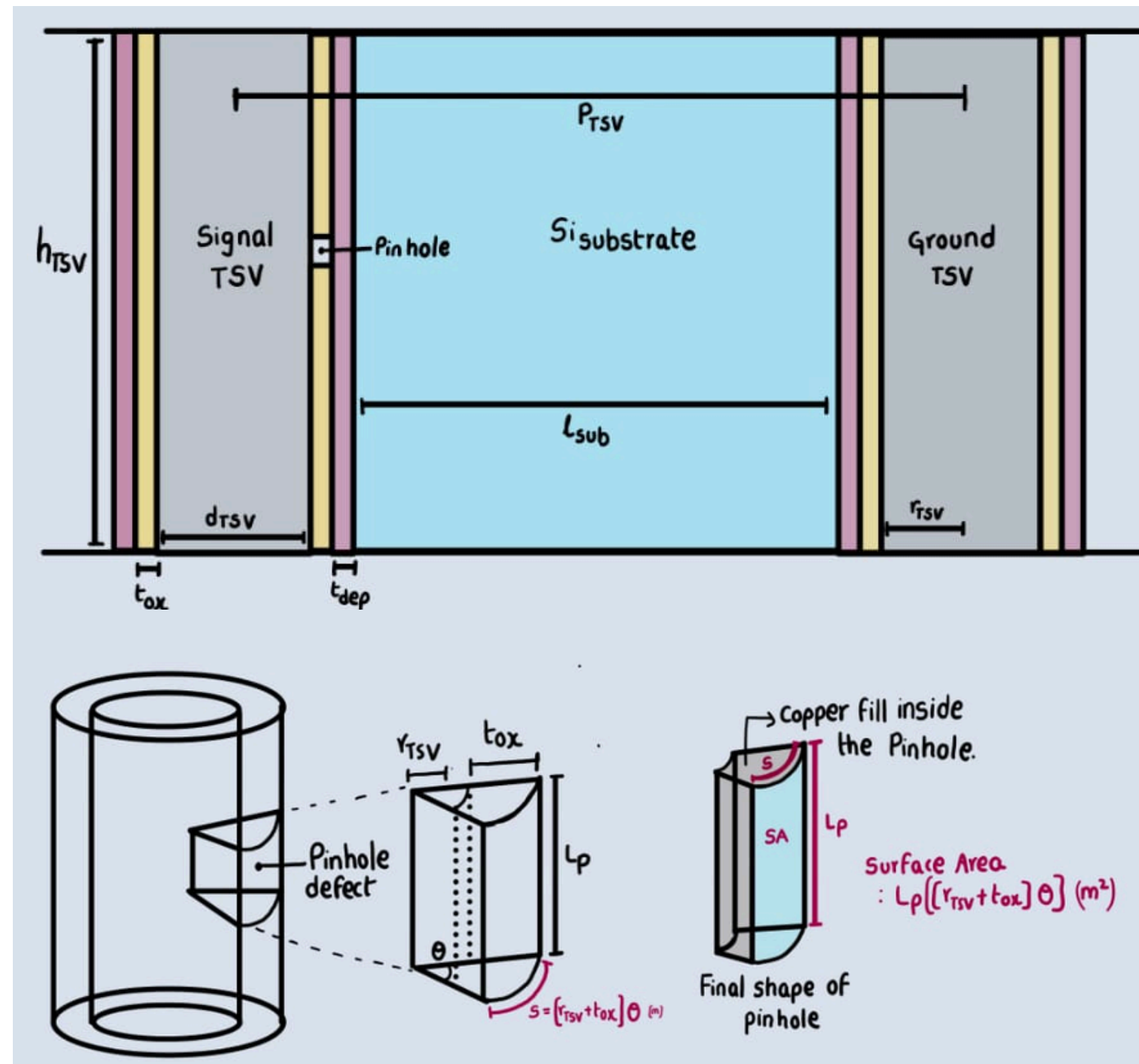


Fig.4.a Defected TSV Physical Structure

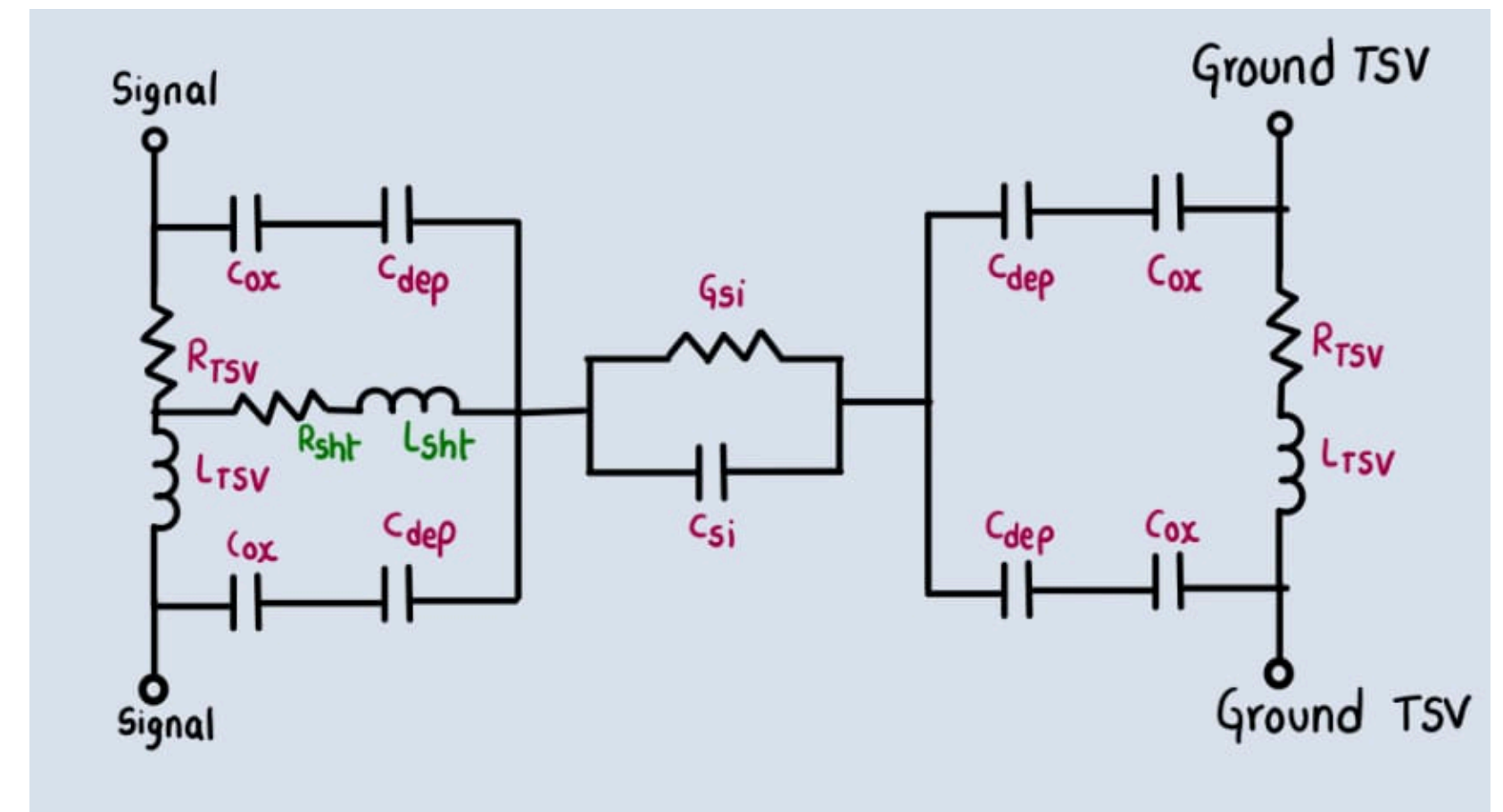


Fig.4.b Equivalent Electrical Circuit of Defected TSV

Pinhole Defected Coupled via (Rsht and Lsht)

Equations

Rtsv, Ltsv, Gsi, Csi, Cox, Cdep equations are the same as Defect Free model

Reference⁽³⁾ Model Equations

$$Rsht = \frac{\rho_T t_{ox}}{Lp^2} [ohms] \quad \dots Eq12$$

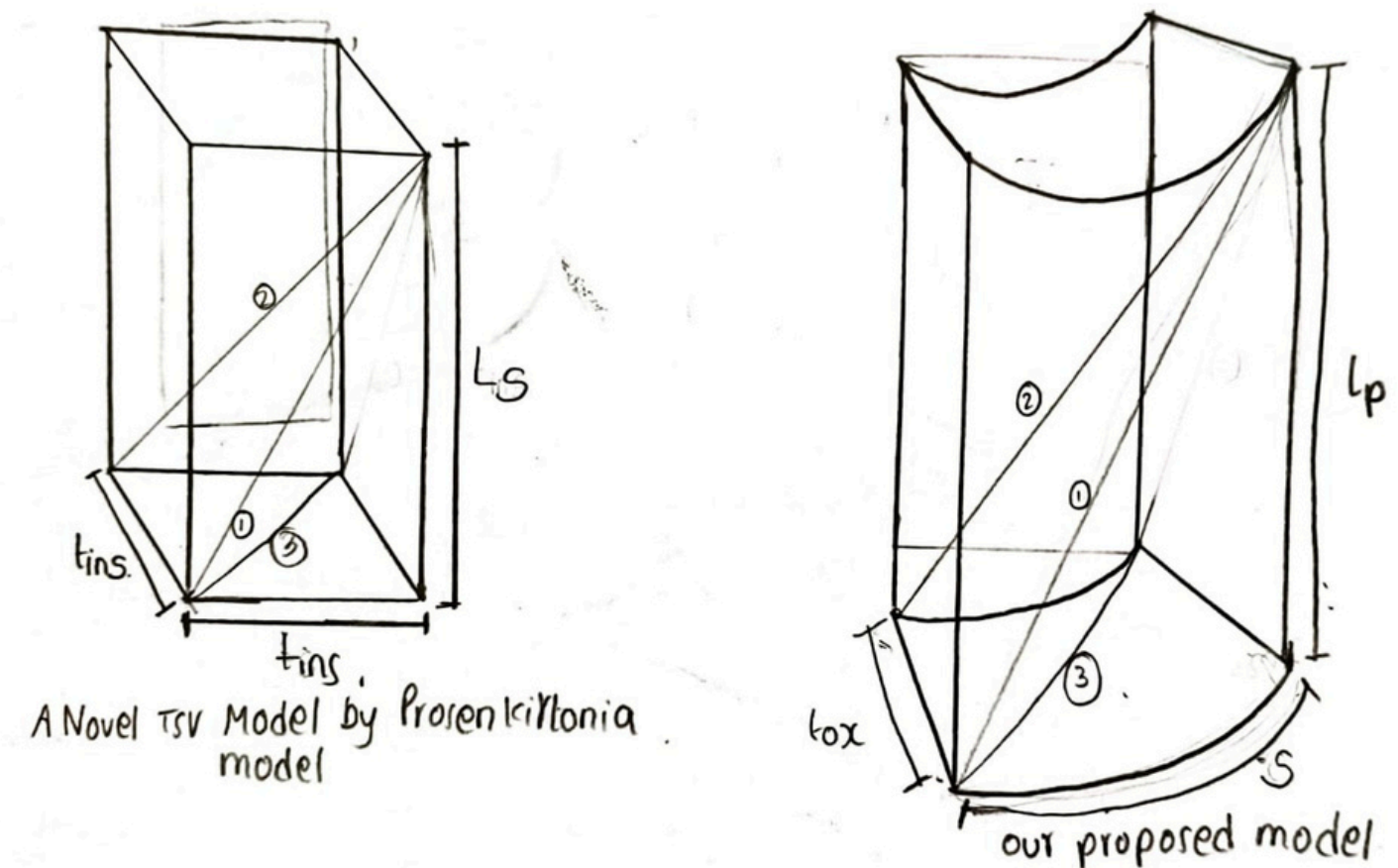


Fig.8 TSV Defected Pinhole Models

$$Lsht = \frac{\mu_{TSV} t_{ox}}{2\pi} \left[\operatorname{arccosh} \left[\frac{tox}{\sqrt{Lp^2 + tox^2}} \right] - \sqrt{2tox^2 + Lp^2} + \frac{tox}{4} + \sqrt{tox^2 + Lp^2} \right] [H] \quad \dots Eq13$$

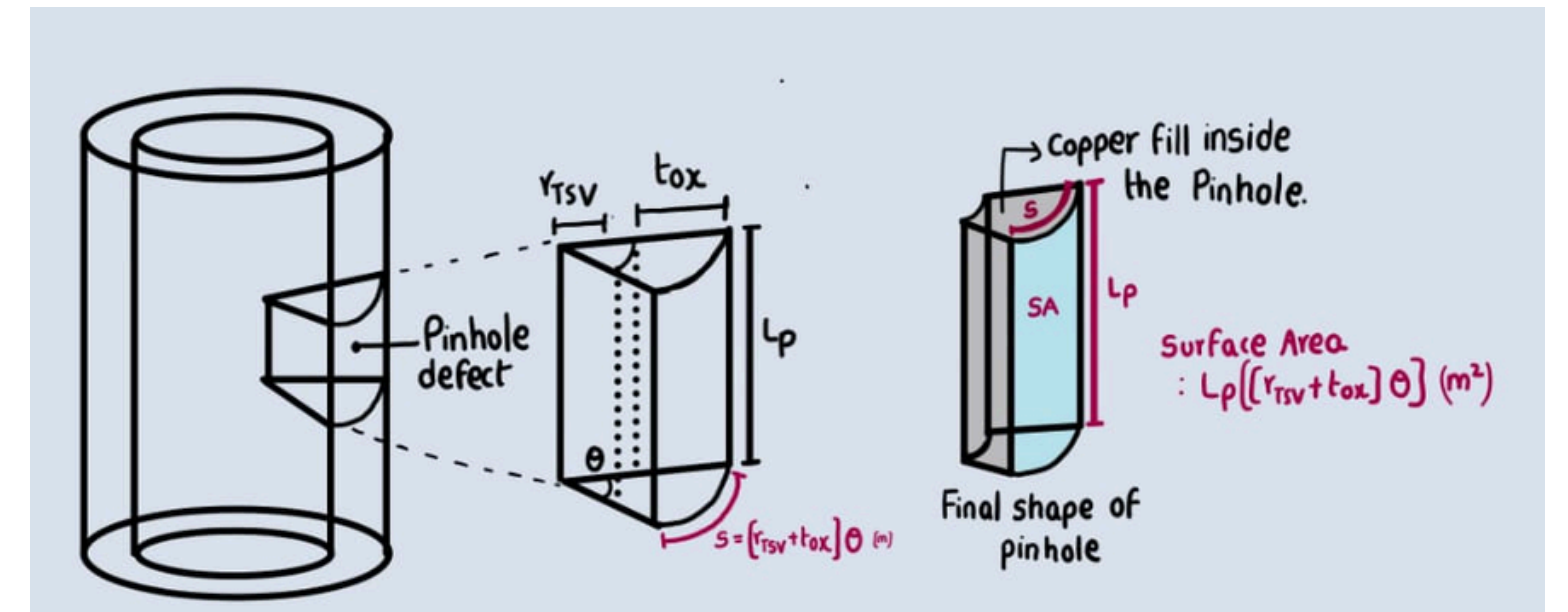
Pinhole Defected Coupled via (Rsht and Lsht)

Equations

Rtsv, Ltsv, Gsi, Csi, Cox, Cdep equations are the same as Defect Free model

Proposed Model Equations

$$Rsht = \frac{\rho_T tox}{Lp [r_{TSV} + tox] \theta} [ohms] \quad \dots Eq14$$



$$Lsht = \frac{\mu_{TSV} t_{ox}}{2\pi} \left[\operatorname{arccosh} \left[\frac{S}{\sqrt{Lp^2 + S^2}} \right] - \sqrt{S^2 + tox^2 + Lp^2} + \frac{S}{4} + \sqrt{S^2 + Lp^2} \right] [H] \quad \dots Eq15$$

Pinhole Defected Coupled via (Rsht and Lsht)

H-Spice Simulation (Paper Model)

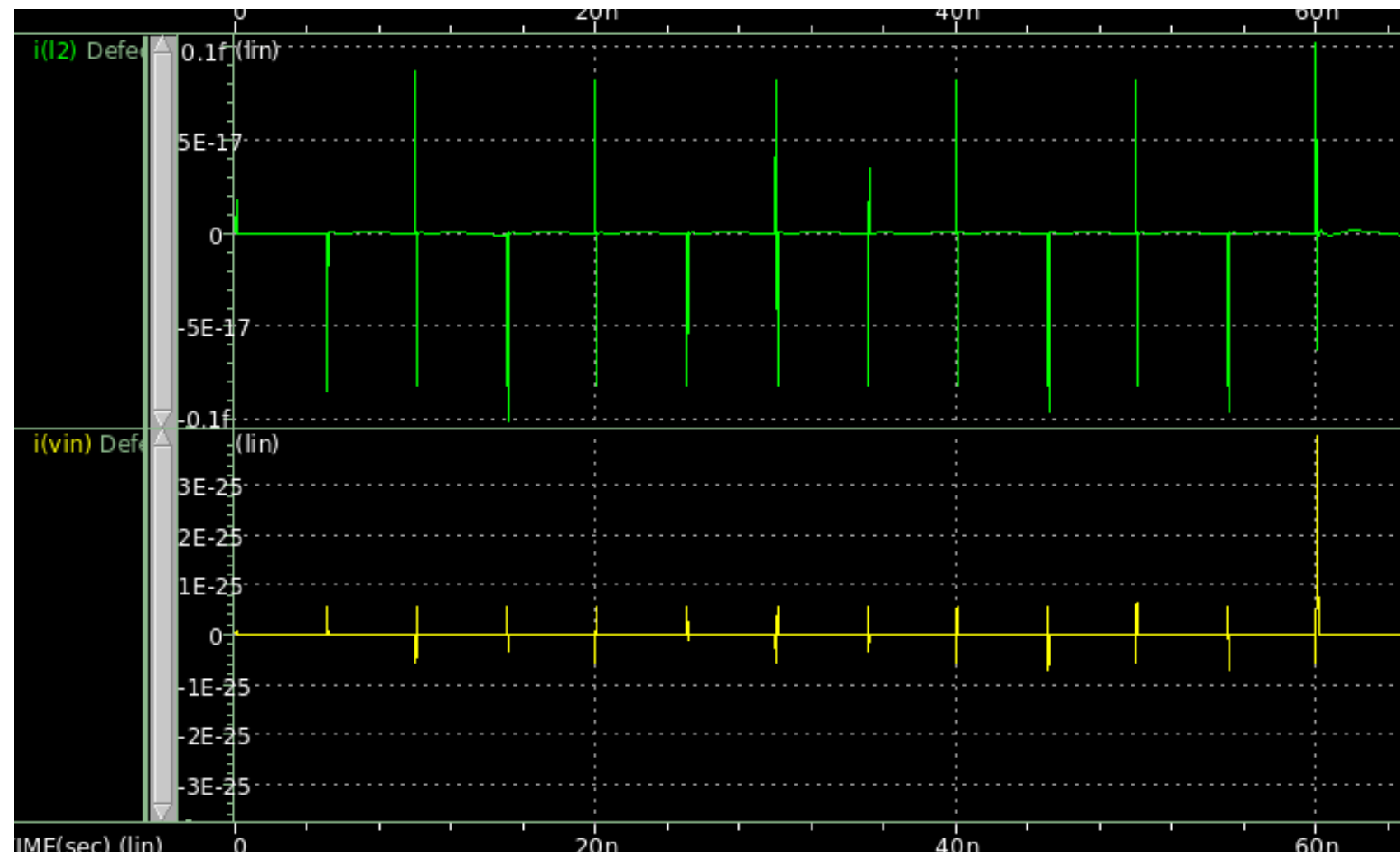


Fig.5.a Reference Paper Model circuit input and output current

H-Spice Simulation (Proposed Model)

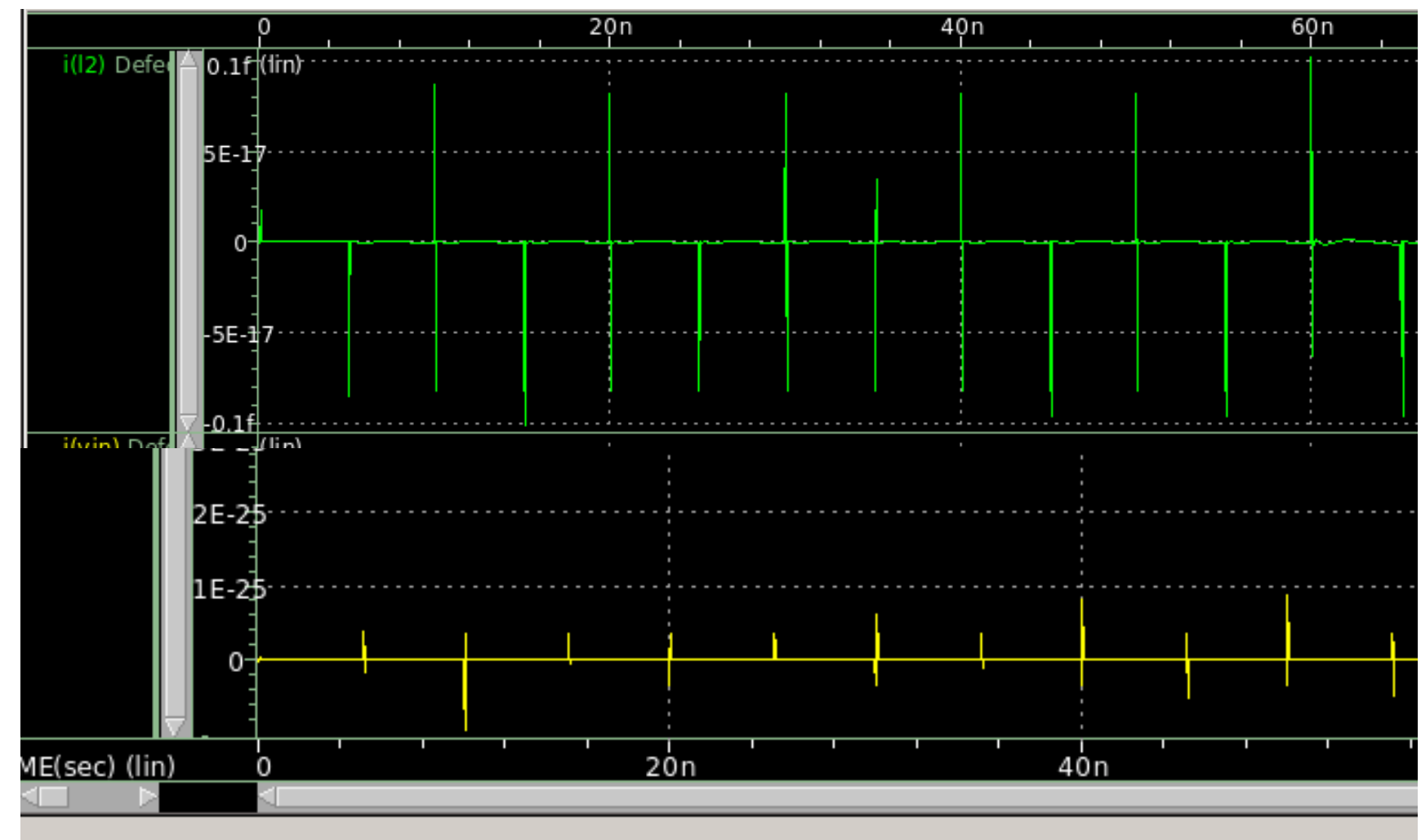


Fig.5.b Proposed Model Circuit input and output current

Pinhole Defected Coupled Via (Cleak and Gleak)

Equivalent Circuit

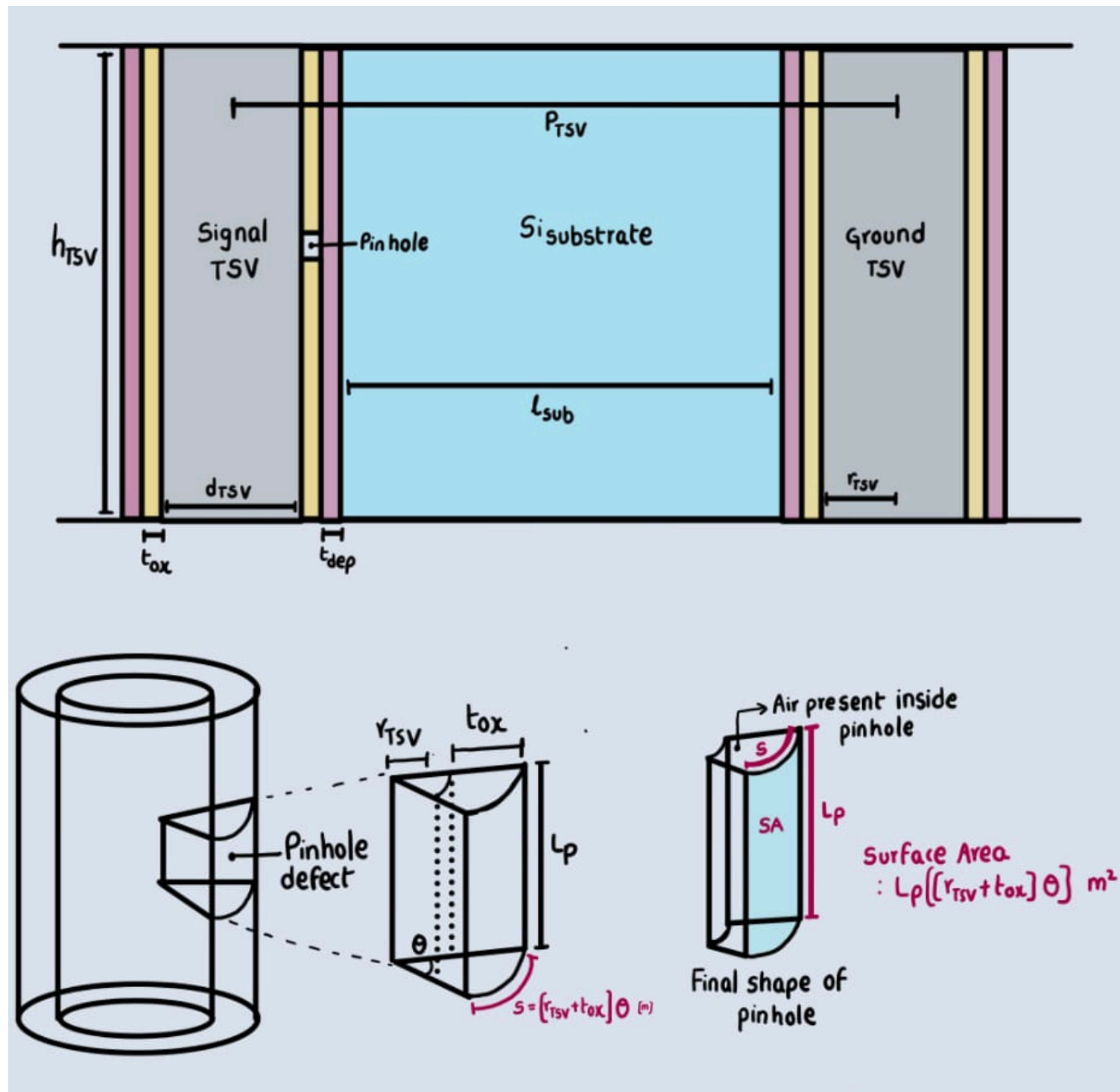


Fig.6.a TSV Defected Structure with Pinhole Cylindrical view

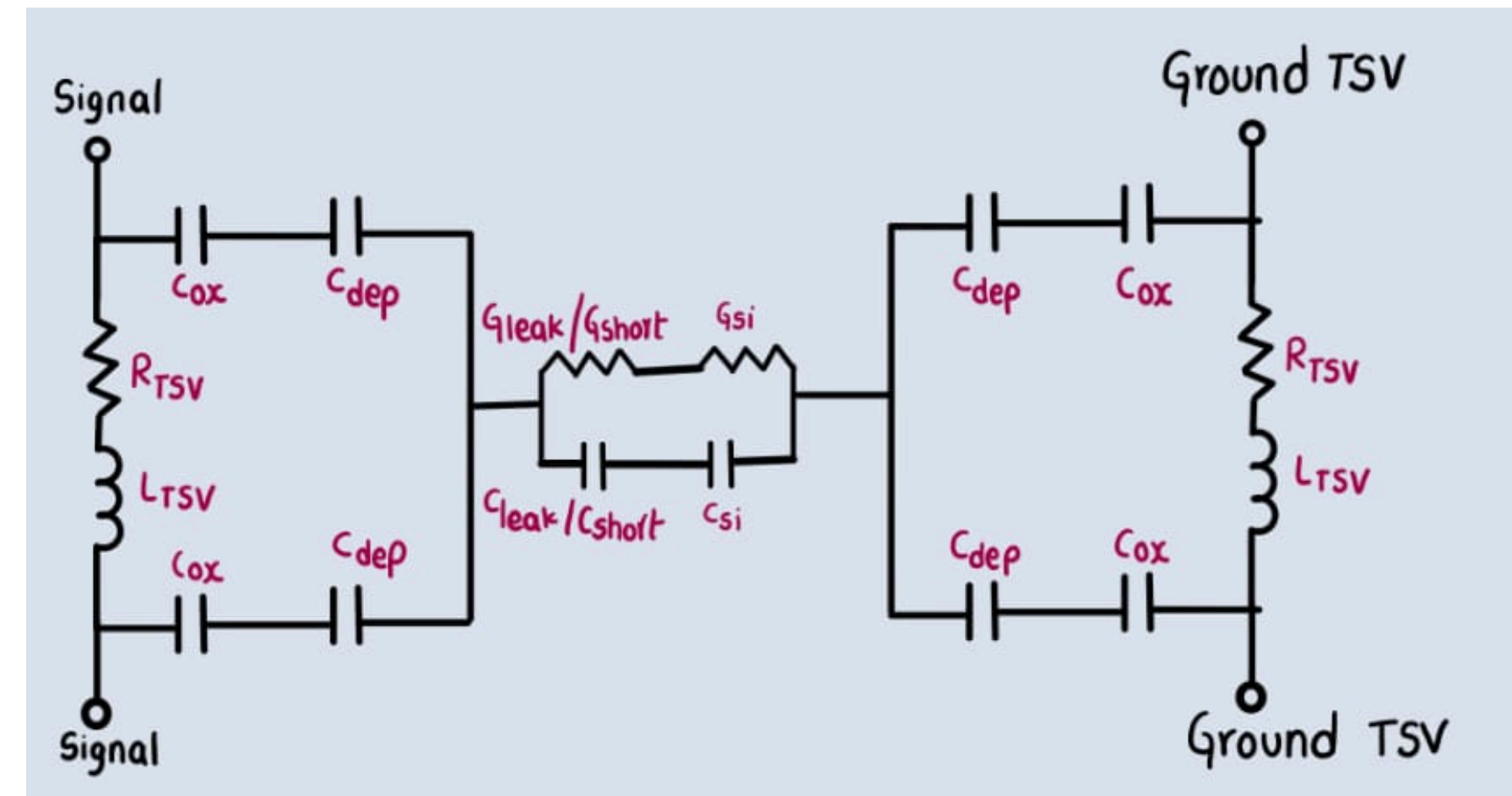


Fig.6.b Equivalent Electrical Circuit of Defected TSV

Pinhole Defected Coupled Via (Cleak and Gleak)

Equations

Rtsv, Ltsv, Cox, Cdep equations are the same as Defect Free model

$$G_{si} = \frac{\pi \sigma_{si} [h_{TSV} - Lp]}{\cosh^{-1} \left[\frac{\rho_{TSV}}{d_{TSV}} \right]} [s] \quad \dots \text{Eq16}$$

$$C_{si} = \frac{\pi \epsilon_{si} [h_{TSV} - Lp]}{\cosh^{-1} \left[\frac{\rho_{TSV}}{d_{TSV}} \right]} [F] \quad \dots \text{Eq17}$$

Reference⁴ Model Equations

$$Cleak = \frac{2\pi \epsilon_{ox} h_{TSV} - 2\pi \epsilon_{air} J r_{TSV}}{\ln \left[\frac{r_{TSV} + tox}{r_{TSV}} \right]} [F] \quad \dots \text{Eq18}$$

$$Gleak = \frac{2\pi \sigma_{ox} h_{TSV} - 4\pi^2 \epsilon_{air} J f r_{TSV}}{\ln \left[\frac{r_{TSV} + tox}{r_{TSV}} \right]} [S] \quad \dots \text{Eq19}$$

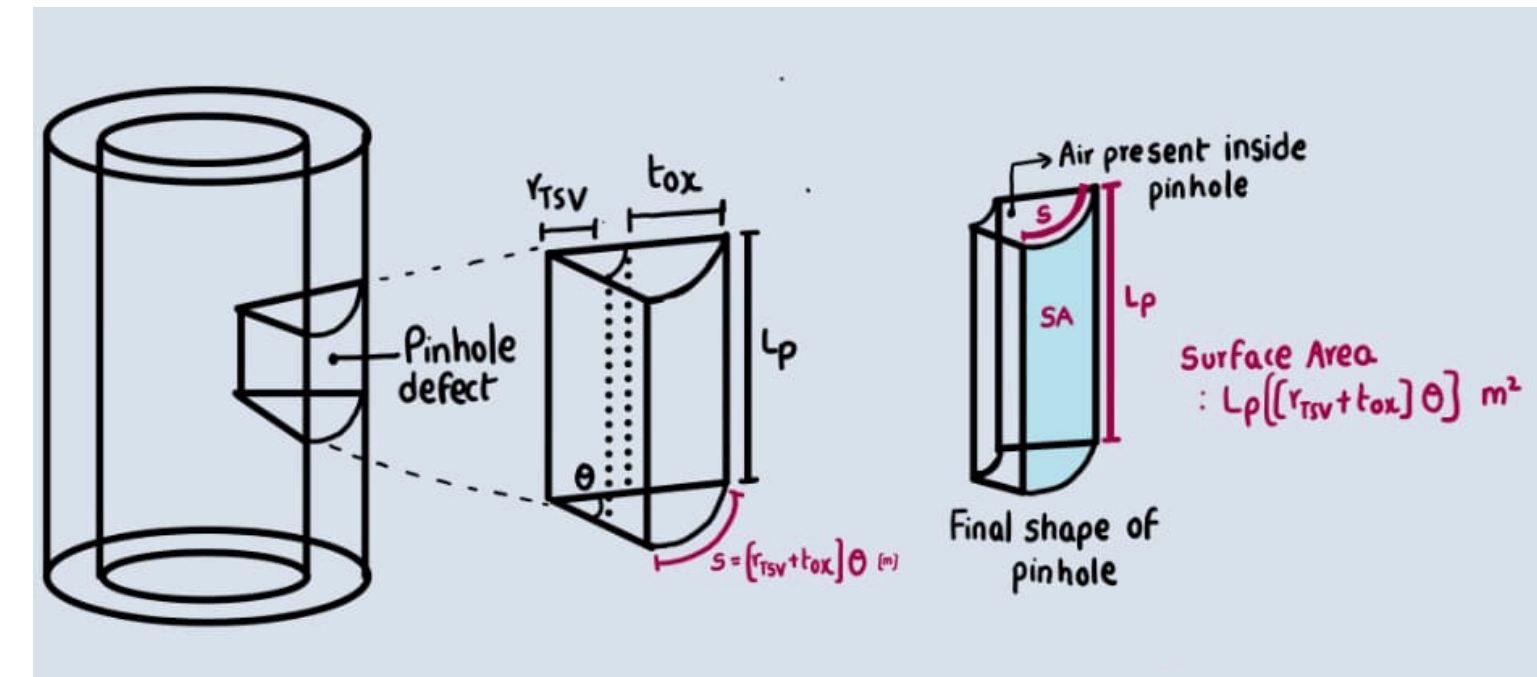


Fig.6.a TSV Defected Pinhole Cylindrical view

Pinhole Defected Coupled Via (Cleak and Gleak)

Equations

Rtsv, Ltsv, Cox, Cdep equations are the same as Defect Free model

$$J = \frac{\theta Lp}{2\pi r_{TSV}} \quad \dots \text{Eq20}$$

Proposed Model Equations

$$C_{short} = \frac{[2\pi - \theta]\epsilon_{ox}Lp + \theta\epsilon_{air}Lp}{\ln \left[\frac{r_{TSV} + tox}{r_{TSV}} \right]} [F] \quad \dots \text{Eq21}$$

$$G_{short} = \frac{[2\pi - \theta]\sigma_{ox}Lp + \theta\epsilon_{air}Lpf}{\ln \left[\frac{r_{TSV} + tox}{r_{TSV}} \right]} [S] \quad \dots \text{Eq22}$$

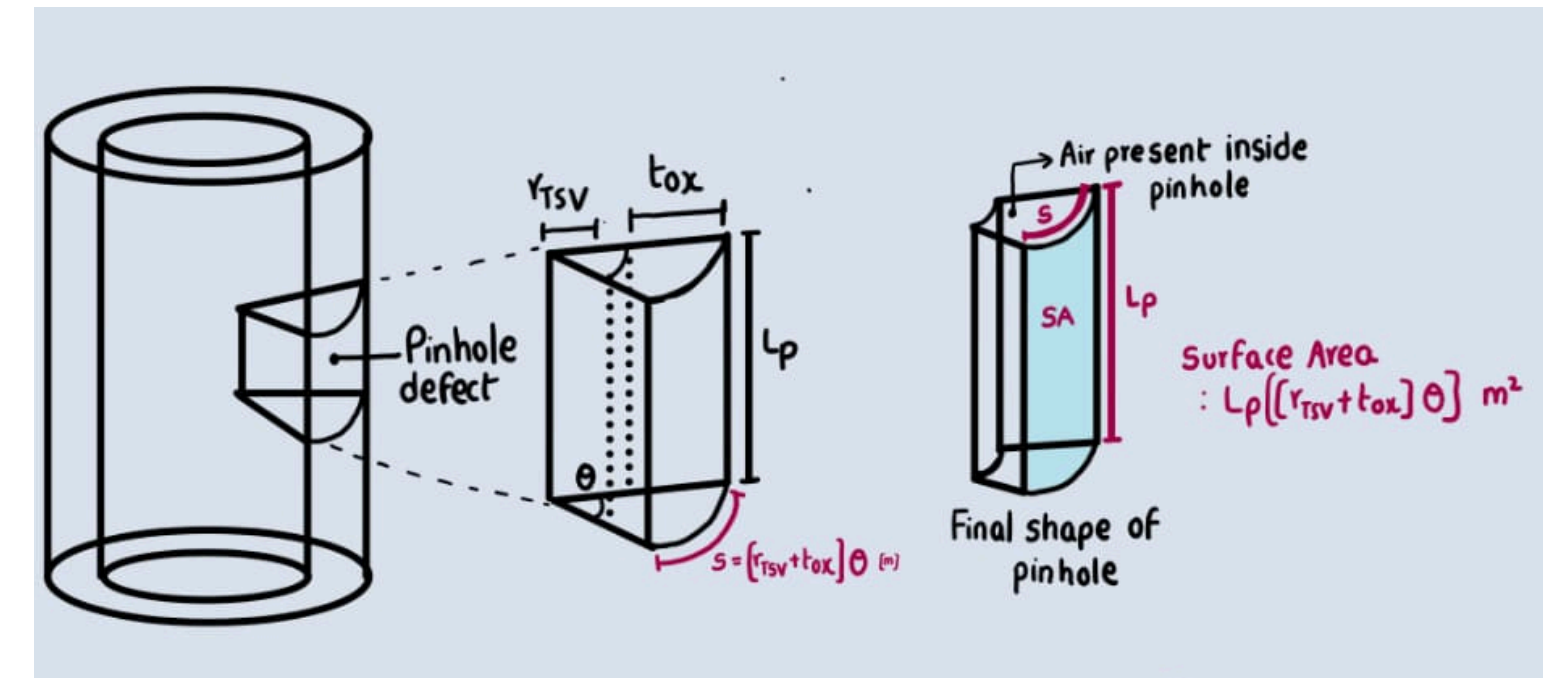


Fig.6.a TSV Defected Pinhole Cylindrical view

Pinhole Defected Coupled Via (Cleak and Gleak)

➡ H-Spcie Simulation
(Paper model)

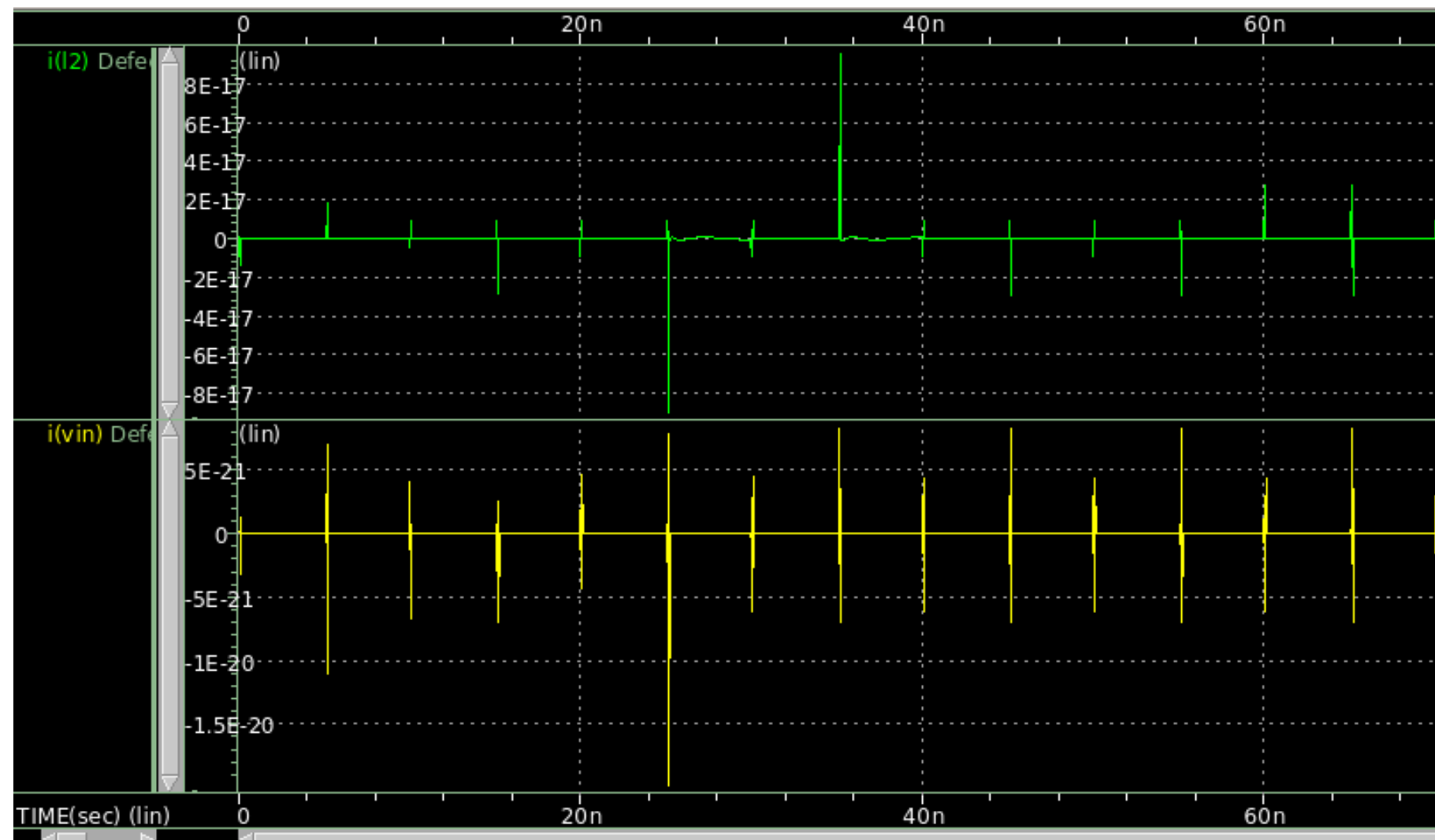


Fig.7.a Reference Model circuit input and output current

➡ H-Spcie Simulation
(Proposed model)

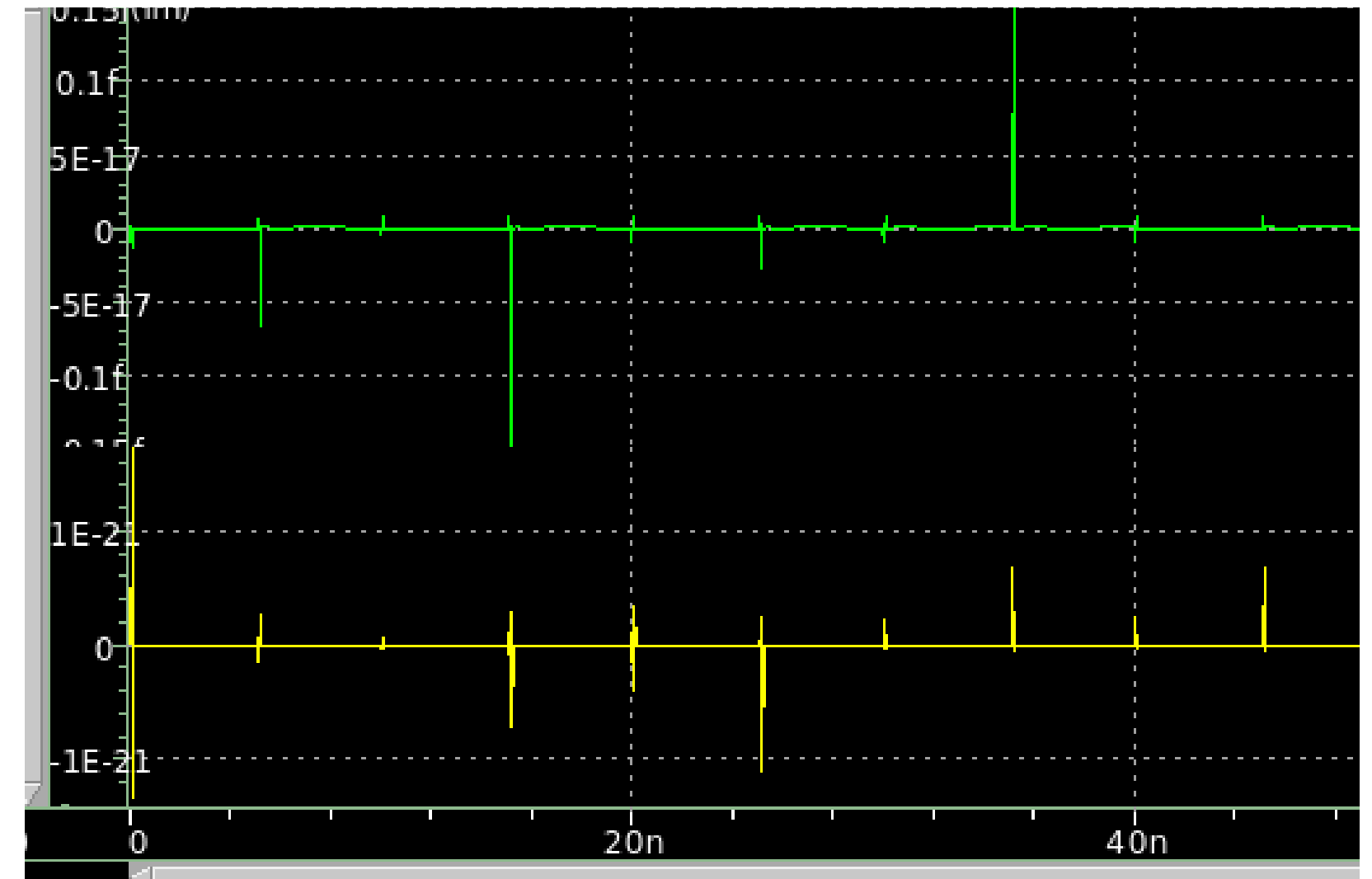


Fig.7.b Proposed Model input and output current

Rise Time

Model Name	Sub Model	Rise Time (s)	Fall Time (s)
Defect-Free TSV	-	3.459e-20	1.454e-20
Defected TSV in terms of R and L	Reference ⁽³⁾ Paper model	1.939e-26	9.926e-24
Defected TSV in terms of R and L	Proposed model	1.928e-26	9.826e-24
Defected TSV in terms of C and G	Reference Paper Model	1.042e-22	2.323e-20
Defected TSV in terms of C and G	Proposed Model	1.623e-22	2.868e-20

Table.3 Rise time and fall time of all TSV Models



References

- ➡ {1} Kim, J., Pak, J. S., Cho, J., Song, E., Cho, J., Kim, H., ... & Kim, J. (2011). High-frequency scalable electrical model and analysis of a through silicon via (TSV). IEEE Transactions on Components, Packaging and Manufacturing Technology, 1(2), 181-195.
- ➡ {2} https://irds.ieee.org/images/files/pdf/2021/2021IRDS_MET.pdf
- ➡ {3} Kirtonia, P., Williams, S., Akter, S., Bayoumi, M., & Khalil, K. (2025). A Novel TSV Model With Fault Characterization for High-Frequency Transmission in 3D ICs. IEEE Transactions on Circuits and Systems I: Regular Papers.
- ➡ {4} Liu, C., Dong, G., Zhi, C., & Zhu, Z. (2025). Modelling and simulation of TSV considering void and leakage defects. Journal of Computational Electronics, 24(2), 61.
- ➡ {5} Grover, F. W. (1946). Inductance Calculations: Working Formulas and Tables. D. Van Nostrand Company, Inc. (Reprinted by Dover Publications in 2004).